

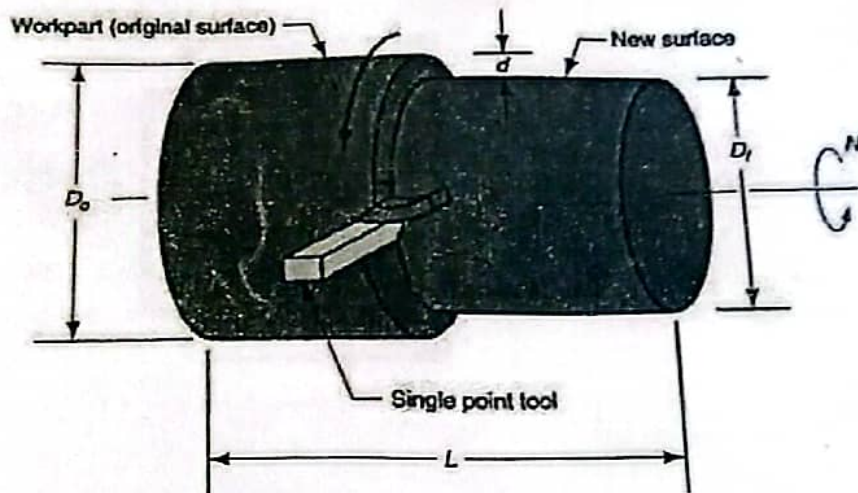
Lathe machine calculations

Turning formulas

1. A cylindrical stainless steel rod with length $L=150$ mm, diameter $D_0 = 12$ mm is being reduced in diameter to $D_f=11$ mm by turning on a lathe. The spindle rotates at $N = 400$ rpm, and the tool is travelling at an axial speed of $v=200$ mm/min

Calculate:

- The cutting speed V (maximum and minimum)
- The material removal rate MRR
- The cutting time t



SOLUTION:

- The maximum cutting speed is at the outer diameter D_0 , and is obtained from the expression

$$V = \pi D_0 N$$

Thus,

$$V_{\max} = (\pi) (12) (400) = 15072 \text{ mm/min}$$

The cutting speed at the inner diameter D_f is

$$V_{\min} = (\pi) (11) (400) = 13816 \text{ mm/min}$$

- From the information given, the depth of cut is

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$$d = (12 - 11) / 2 = 0,5 \text{ mm}$$

and the feed is

$$f = v / N$$

$$f = 200 / 400 = 0,5 \text{ mm/rev}$$

thus the material removal rate is calculated as

$$MRR = (\pi) (D_{avg}) (d) (f) (N) = (\pi) (11,5) (0,5) (0,5) (400) = 3611 \text{ mm}^3/\text{min} = 60,2 \text{ mm}^3/\text{s}$$

c. The cutting time is

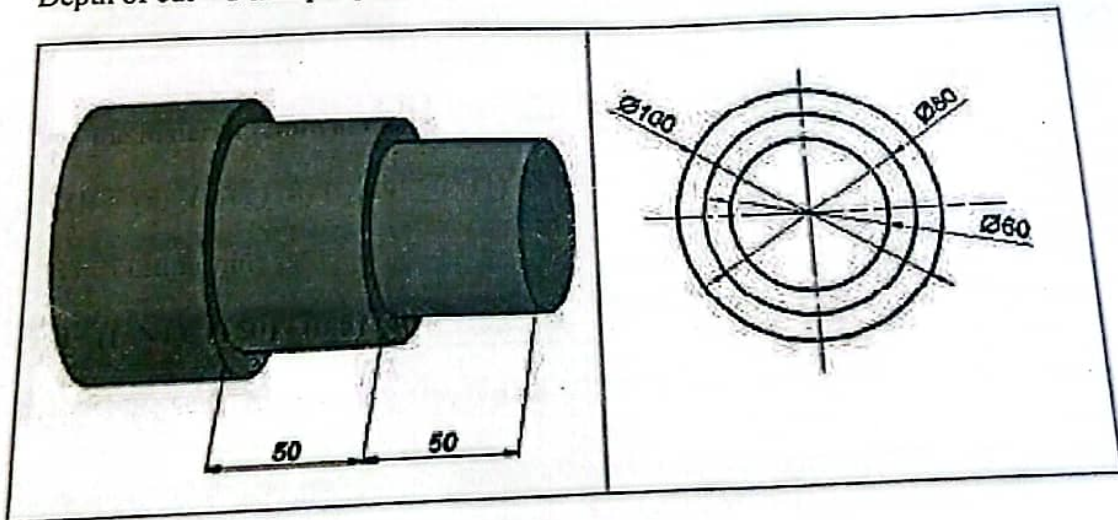
$$t = l / (f \cdot N) = (150) / (0,5) (400) = 0,75 \text{ min}$$

2. The part shown below will be turned in two machining steps. In the first step a length of $(50 + 50) = 100 \text{ mm}$ will be reduced from $\varnothing 100 \text{ mm}$ to $\varnothing 80 \text{ mm}$ and in the second step a length of 50 mm will be reduced from $\varnothing 80 \text{ mm}$ to $\varnothing 60 \text{ mm}$. Calculate the required total machining time T with the following cutting conditions:

Cutting speed $V=80 \text{ m/min}$,

Feed is $f=0.8 \text{ mm/rev}$,

Depth of cut = 3 mm per pass.



SOLUTION:

$$V=80 \text{ m/min}$$

$$f=0.8 \text{ mm/rev}$$

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The turning will be done in 2 steps. In first step a length of $(50 + 50) = 100$ mm will be reduced from $\varnothing 100$ mm to $\varnothing 80$ mm and in second step a length of 50 mm will be reduced from $\varnothing 80$ mm to $\varnothing 60$ mm.

Step 1

$\varnothing 100$ mm to $\varnothing 80$ mm a length of 100mm

$$N = \frac{1000 \cdot V}{\pi \cdot D_o} = \frac{1000 \cdot 80}{3.14 \cdot 100} = 255 \text{ rpm}$$

$$\text{Number of pass} = n_p = \frac{\text{Depth of material to be removed}}{\text{Depth of cut}} = \frac{(100 - 80)}{2 \cdot 3} = 4$$

$$\text{Time required} = T_c = \frac{L}{f \cdot N} \cdot n_p = \frac{100}{0.8 \cdot 255} \cdot 4 = 1.96 \text{ min.}$$

Step 2

To turn from $\varnothing 80$ to $\varnothing 60$ for 50mm long.

$$N = \frac{1000 \cdot V}{\pi \cdot D_o} = \frac{1000 \cdot 80}{3.14 \cdot 80} = 318 \text{ rpm}$$

$$\text{Number of pass} = n_p = \frac{\text{Depth of material to be removed}}{\text{Depth of cut}} = \frac{(80 - 60)}{2 \cdot 3} = 4$$

$$\text{Time required} = T_c = \frac{L}{f \cdot N} \cdot n_p = \frac{50}{0.8 \cdot 318} \cdot 4 = 0.79 \text{ min}$$

$$\text{Total time} = 1.96 + 0.79 = 2.75 \text{ min.}$$

3. A mild steel rod having 50 mm diameter and 500 mm length is to be turned on a lathe. Determine the machining time to reduce the rod to 45 mm in one pass when cutting speed is 30 m/min and a feed of 0.7 mm/rev is used.

Solution

Given data: $D_o = 50$ mm, $L = 500$ mm

$v = 30$ m/min, $f = 0.7$ mm/rev

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Substituting the values of v and D_0 in $V = \pi D_0 N / 1000 \text{ M/min}$

Required spindle speed as: $N = 191 \text{ rpm}$

$$T_c = \frac{L}{f \cdot N} = \frac{500}{0.7 \cdot 191} = 3.73 \text{ min.}$$

4. Determine the angle at which the compound rest would be swiveled for cutting a taper on a work piece having a length of 150 mm and outside diameter 80 mm. The smallest diameter on the tapered end of the rod should be 50 mm and the required length of the tapered portion is 80 mm.

Solution

Given data: $D_0 = 80 \text{ mm}$, $D_f = 50 \text{ mm}$, $L = 80 \text{ mm}$ (with usual notations)

$$\tan \alpha = \frac{D_0 - D_f}{2 \cdot L}$$

$$\tan \alpha = (80 - 50) / 2 \cdot 80 \quad \text{or} \quad \alpha = 10.620^\circ$$

The compound rest should be swiveled at 10.62°

5. The work piece has 150mm long and 50mm diameter turning on a lathe reduces to 30mm long and 30mm diameter. The work piece at the cutting speed of 80m/min, the feed is 0.2mm/rev and maximum depth of cut is 3mm. Calculate the total time required to machine a work piece? Assume Approach and over travel distance are 3mm and 3mm respectively, for turning operation.

Solution:

Time for Turning:

Total length of tool travel = job length + length of approach and over travel

$$L = 150 + 3 + 3 = 156 \text{ mm}$$

$$\text{Required depth to be cut} = (50 - 30) / 2 = 10 \text{ mm}$$

Since maximum depth of cut is 3 mm, 10 mm cannot be cut in one pass. Therefore, we calculate number of cuts or passes required.

$$\text{Number of cuts required} = 10 / 3 = 3.33 \text{ or } 4 \text{ (since cuts cannot be a fraction)}$$

$$\text{Machining time for one cut} = L / (f \times N)$$

$$N = KV / 3.14 \cdot D_0 = 1000 \cdot 80 / 3.14 \cdot 80 = 318.47 \text{ rpm}$$

$$\text{Total turning time} = [L / (f \times N)] \times \text{Number of pass}$$

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$$=(156/0.2 \times 318.47) \times 4$$

$$=9.79 \text{ min.}$$

Time for facing: Now, the diameter of the job is reduced to 50 mm. Recall that in case of facing operations, length of tool travel is equal to half the diameter of the job. That is, $L = 15 \text{ mm}$. Substituting in equation, we get

$$T_c = 15 / (0.2 \times 318.47)$$

$$= 0.23 \text{ min.}$$

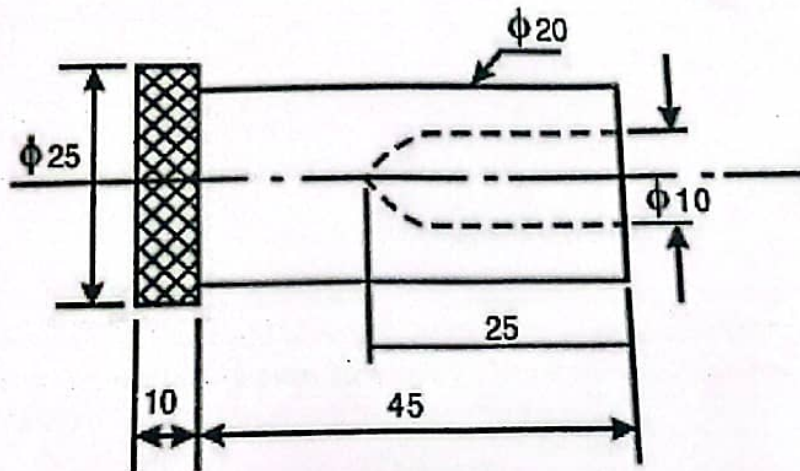
Total time:

Total time for machining = Time for Turning + Time for Facing

$$= 9.79 + 0.23$$

$$= 10.02 \text{ min.}$$

6. The shaft shown in the figure below is to be machined on a lathe from a $\phi 25 \text{ mm}$ bar. Calculate the machining time if speed V is 60 m/min. , turning feed is 0.2 mm/rev , drilling feed is 0.08 mm/rev and knurling feed is 0.3 mm/rev .



Solution:

Step 1

Facing $\phi 25 \text{ mm}$ bar on both side because multiple set up

$$\text{Length of cut} = L = D/2 = 25/2 = 12.5 \text{ mm}$$

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$$N = \frac{1000 \cdot 60}{25 \cdot \pi} = 764 \text{ rpm}$$

$$T_c = \frac{L}{f \cdot N} = \frac{12.5}{0.2 \cdot 764} = 0.082 \text{ min}$$

Time to facing on both ends = $2 \cdot 0.082 = 0.164 \text{ min}$.

Step 2

Turn $\varnothing 25 \text{ mm}$ to $\varnothing 20 \text{ mm}$

$$T_c = \frac{L}{f \cdot N} = \frac{45}{0.2 \cdot 764} = 0.29 \text{ min}$$

Step 3.

Drilling of 8mm diameter hole

$$N = \frac{1000 \cdot 60}{10 \cdot \pi} = 1910 \text{ rpm}$$

$$T_c = \frac{L}{f \cdot N} = \frac{25}{0.08 \cdot 1910} = 0.16 \text{ min}$$

Step 4

$$N = \frac{1000 \cdot 60}{25 \cdot \pi} = 764 \text{ rpm}$$

$$T_c = \frac{L}{f \cdot N} = \frac{10}{0.3 \cdot 764} = 0.04 \text{ min}$$

Total machining time = $T_{c1} + T_{c2} + T_{c3} + T_{c4}$

$$= 0.164 + 0.29 + 0.16 + 0.04 = 0.65 \text{ min}.$$

7. A batch of 800 workpieces is to be produced on a turning machine. Each workpiece with length $L=120 \text{ mm}$ and diameter $D=10 \text{ mm}$ is to be machined from a raw material of $L=120 \text{ mm}$ and $D=12 \text{ mm}$ using a cutting speed $V=32 \text{ m/min}$ and a feed rate $f=0.8 \text{ mm/rev}$. How many times we have to sharpen or regrind the cutting tool?

In the Taylor's expression, use constants as $n=1.25$ and $C=175$.

Solution:

Given

$L=120 \text{ mm}$, $D_0=12 \text{ mm}$, $D_f=10 \text{ mm}$; $V=32 \text{ m/min}$; $f=0.8 \text{ mm/rev}$, $n=1.25$, $C=175$

No of sharpening required:

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From Taylors tool life equation, we have

$$VT^n = C$$

$$32 \cdot T^{1.25} = 175$$

$$\left(\frac{175}{32}\right)^{1/1.25}$$

$$T = 3.89 \text{ min.}$$

$$\text{Also, } V = \frac{\pi \cdot D \cdot N}{1000},$$

$$N = \frac{32 \cdot 1000}{3.14 \cdot 12} = 849 \text{ rpm}$$

$$\text{Now, machining time per piece} = \frac{L}{f \cdot N} = \frac{120}{0.8 \cdot 849} = 0.177 \text{ min}$$

$$\text{Therefore, machining time for 800 work piece} = 800 \cdot 0.177 = 141.6 \text{ min.}$$

$$\text{Hence, Number of resharpenings required} = \frac{T_c}{T_l} = 36.4 \text{ Or } 36 \text{ resharpenings}$$